

BISHESH RAJ KHANAL

AI-driven Backend and Platform Engineer

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🌐 Bishesh-Khanal

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🔗 Portfolio

👤 Professional Summary

High-ownership **AI-Integrated Platform and Backend Engineer** with proven expertise in building secure, distributed, end-to-end microservices and spearheading enterprise-grade observability. Skilled in designing and shipping bulletproof integration platforms handling third-party webhooks, payments, and notifications with rigorous security standards to protect revenue and customer trust. Comfortable with deep architectural knowledge spanning Domain-Driven Design (DDD), monorepos, and containerized microservice ecosystems. Passionate about bridging robust backend reliability with modern AI-driven integrations to solve complex scalability problems.

💻 Technical Skills

🔧 Core Backend and Integration

Python, C++, C, C#, Rust, Stripe, Twilio, Sendgrid, RabbitMQ, Keycloak

🗄 Database and ORM

PostgreSQL, CockroachDB, Alembic, Redis, ByteBase, DBeaver, PgAdmin

📷 Observability and Reliability

OpenTelemetry (OTeL), OpenObserve, Distributed Tracing, Metrics and Logs, SLO Configuration

🔒 Security

Cryptography, Email Security

🏠 Architecture

Monolithic, Microservice, Monorepo Setup, Domain Driven Design (DDD), Webhook-driven systems

⚙ DevOps

GitHub Actions, Docker, EKS, Kubernetes, EC2, Argo CD, Git, CMake, GDB

🧠 AI/ML and Data

FastAI, Scikit-learn, PyTorch, PyTorch Geometric, Vector and Graph Data Processing, BeautifulSoup, Selenium, S3, MinIO, Power BI

🤖 Automation

n8n, Zapier, Agentic Automation, Non-Agentic Automation, Workflow Automation

🏢 Work Experience

Backend and Platform Engineer

October, 2025 - June, 2026

Kyra Works Pvt. Ltd.

Banepa, Nepal

- **Owned and Architected Revenue-Critical Backend Microservices:** Built, launched, and maintained scalable billing and notification microservices while integrating **Stripe**, **Twilio**, and **Sendgrid** into the company's core SaaS product, directly enabling payment processing and customer communication at scale.
- **Hardened Webhook and System Security:** Designed and implemented secure webhook processing pipelines featuring signature header verification, timestamp validation, and graceful fallback mechanisms to safeguard high-throughput payment and messaging data against replay attacks and service drops.
- **Configured Email Deliverability Infrastructure:** Managed enterprise email authentication pipelines by

configuring and validating **SPF**, **DKIM**, and **DMARC** policies, significantly lowering spam placement and ensuring high transactional email deliverability.

- **Initiated Enterprise Observability Platform:** Spearheaded the company-wide deployment of **OpenTelemetry (OTel) Collectors** to aggregate distributed traces, logs, and metrics across Python microservices, **RabbitMQ**, **Keycloak**, and **Redis**, pushing telemetry data to **OpenObserve** to establish measurable **Service Level Objectives (SLOs)**.
- **Streamlined Enterprise Workflows:** Engineered complex event-driven and AI-agentic automation workflows using **n8n** and **Zapier**, incorporating LLM-based decision steps for dynamic routing and data enrichment, reducing manual intervention and bridging cross-platform data synchronization.
- **Standardized Infrastructure and Delivery:** Helped build a cloud-native delivery pipeline using GitHub Actions for CI, Docker for containerization, Amazon EC2 for compute infrastructure, Kubernetes (Amazon EKS) for orchestration, and Argo CD for GitOps-based deployments.

***Note on Transition:** Following the successful completion of my contract, I chose not to extend it in order to dedicate time to travel, personal software engineering projects, and completing my degree requirements.*

Selected Projects

Fake News Detector (Python, Scikit-learn, NLP)

- Built a Python classification model for real-world news data, presented it at an **International AI conference in India** and won the **Best Paper Presentation Award** among **80 participants**.

Novel Fake News Detection using Graph Neural Networks (GNN) (Python, PyTorch Geometric, NetworkX)

- Designed a graph neural network pipeline modeling news propagation, converting unstructured text into graph representations for training; applied research bridging NLP and graph deep learning.

Automated Job Market Data Pipeline (Python, BeautifulSoup, Requests, Panel)

- Engineered a resilient web scraping and data extraction pipeline to gather, clean, and structure market listings, storing analytical metrics to generate dynamic business decision-making dashboards.

C++ Systems and Pathfinding Engine (C++, SFML, Algorithms)

- Engineered high-performance C++ systems featuring dynamic vision-blocking, custom dynamic lighting solvers, and graph path-finding algorithms focused on deterministic performance and memory efficiency.

Achievements and Certificates

Best Paper Presentation Certificate at the DST SERB-sponsored International Conference on Artificial Intelligence of Things for Sustainability, held in India on January 20th and 21st, 2024 (AIoT4S 2024)

Education

Bachelor of Science in Computer Science, Kathmandu University

August, 2021 - July, 2026

Relevant Coursework: Artificial Intelligence (AI), Data Structures & Algorithms, Computer Architecture, Object-Oriented Programming (OOP), Database Management Systems, Compiler Design, Microprocessor